Traveling Salesman Problem

Classically, the TSP deals with finding the shortest (closed) tour in an n-city situation, where each city is visited exactly once before returning back to the starting point. The associated TSP model is defined by two pieces of data:

The number of cities n,

The distance d_{ij} between cities i and j ($d_{ij}=\infty$ if cities i and j are not linked).

The maximum number of tours in an n-city situation is (n-1)! if the network is directed (i.e. $d_{ij}\neq d_{ji}$) and half that much if it is not.

Model:
The definition of a tour disallows linking a city to itself by assigning a very high penalty to the diagonal elements of the distance matrix. A TSP is symmetric if $d_{ij}=d_{ji}$ for all i and j; else it is asymmetric.

Define $x_{ij}=1$ if city j is reached from city i, $x_{ij}=0$ otherwise

The TSP model is given as:

$$\text{minimize} \quad z = \sum_{i=1}^{n}\sum_{j=1}^{n} d_{ij}\cdot x_{ij} \quad , d_{ij}=\infty \text{ for all } i=j$$

$$\text{s.t} \quad \sum_{j=1}^{n} x_{ij} = 1, \quad i=1,2,\dots,n \qquad (1)$$

$$\sum_{i=1}^{n} x_{ij} = 1, \quad j=1,2,\dots,n \qquad (2)$$

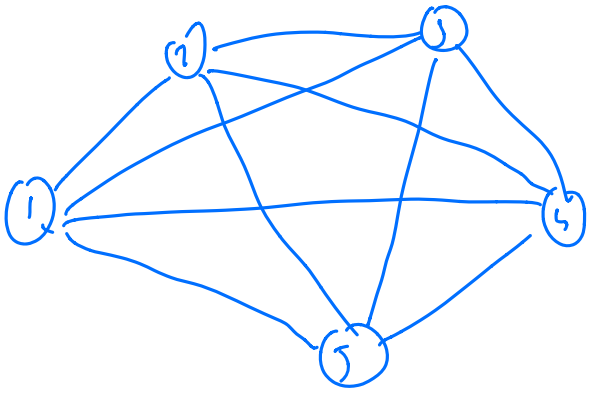
$$x_{ij} \in \{0,1\}. \qquad (3)$$

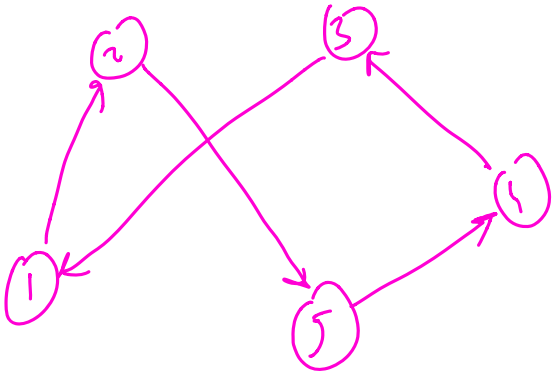
Solution forms an n-city tour. (4)

Linear assignment problem

Constraints (1), (2) and (3) define a regular assignment model in which $x_{ij}=1$ if node i is linked to node j, and zero otherwise. If the solution of the assignment model happens to be a tour (i.e. it satisfies constraint (4)), then it is automatically optimal for the TSP. However, the assignment model is likely to consist of subtours. Additional computations are then needed to determine the optimal solution.

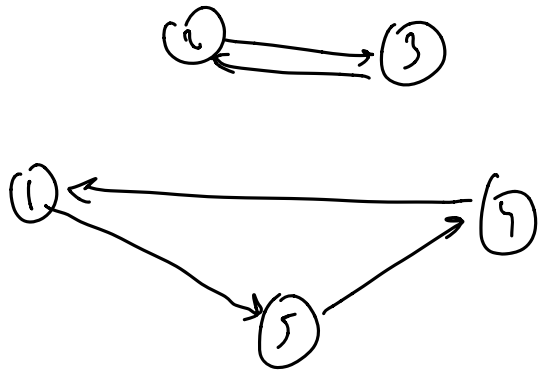
Ex:





$x_{12} = x_{25} = x_{54} = x_{43} = x_{31} = 1$

Tour solution



$x_{23} = x_{32} = x_{15} = x_{54} = x_{41} = 1$

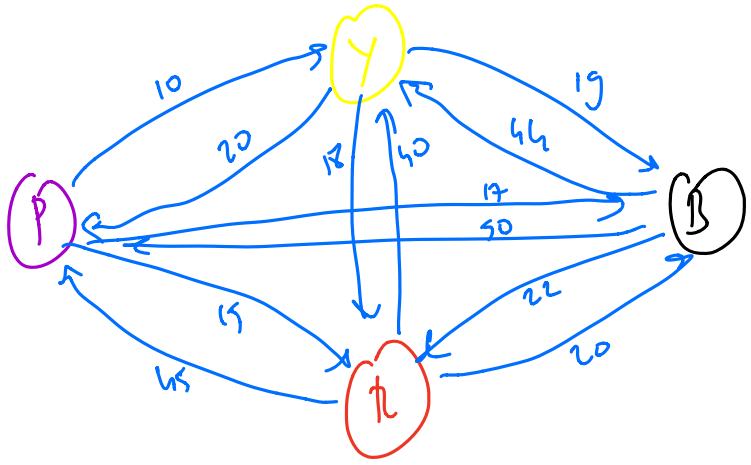
subtour solution !

Ex:

The daily production schedule at the Rainbow Company includes batches of purple (P), yellow (Y), red (R), and black (B) paints. The production facility must be cleaned between successive batches. Table below summarizes the cleanup times in minutes. The objective is to determine the sequencing of colors that minimizes the total cleanup time.

	Interbatch cleanup time (min)			
Paint	Purple	Yellow	Black	Red
Purple	∞	10	17	15
Yellow	20	∞	19	18
Black	50	44	∞	22
Red	45	40	20	∞

Note that the matrix is not symmetric.



In the TSP model, each color represents a city, and the cleanup time between two successive colors represents distance. Let M be sufficiently large penalty and define

x_{ij}=1 if paint j follows paint i and zero otherwise.

The TSP model is given as:

$$\text{minimize } z = 10x_{PY} + 17x_{PB} + 15x_{PR} + 20x_{YP} + 19x_{YB} + 18x_{YR} + 50x_{BP} + 45x_{BY} + 25x_{BR} + 45x_{RP} + 40x_{RY} + 20x_{RB} + M.x_{PP} + M.x_{YY} + M.x_{BB} + M.x_{RR}$$

s.t

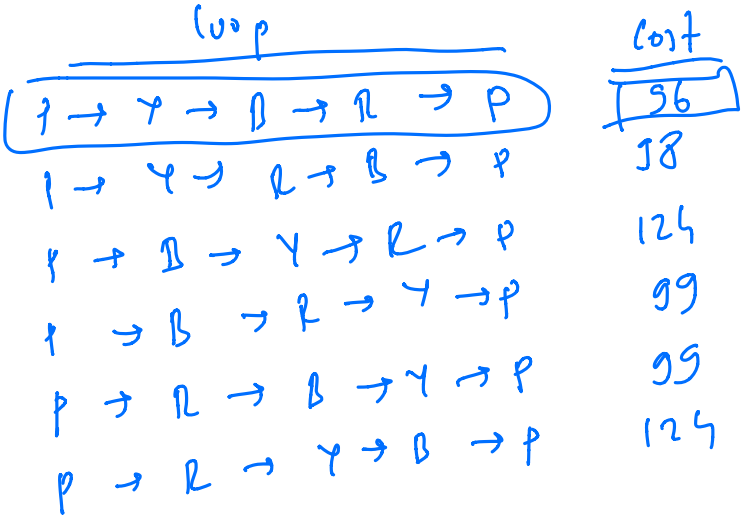
$$x_{PP} + x_{PY} + x_{PB} + x_{PR} = 1$$
$$x_{YP} + x_{YY} + x_{YB} + x_{YR} = 1$$
$$x_{BP} + x_{BY} + x_{BB} + x_{BR} = 1$$
$$x_{RP} + x_{RY} + x_{RB} + x_{RR} = 1$$

$$x_{PP} + x_{YP} + x_{BP} + x_{RP} = 1$$
$$x_{PY} + x_{YY} + x_{BY} + x_{RY} = 1$$
$$x_{PB} + x_{YB} + x_{BB} + x_{RB} = 1$$
$$x_{PR} + x_{YR} + x_{BR} + x_{RR} = 1$$
$$x_{ij} \in \{0,1\}$$

Solution is a tour.

The use of penalty M in the objective function is equivalent to deleting x_{pp}, x_{yy}, x_{bb} and x_{rr} from the model.

A straightforward way to solve TSP is exhaustive enumeration. The maximum number of tours in an n-city problem is (n-1)!. For the present example, number of possible tours is 6. Following table lists and evaluates all six tours and shows that W->Y->B->R->W is optimum.



Exhaustive enumeration is not practical for the general TSP. Instead, we can use one of two exact algorithms: B&B and cutting plane. However, these methods are not computationally reliable. For this reason heuristics are used to obtain good (but not necessarily optimal) solutions to the problem. Three of these are simulated annealing, genetic algorithms and migrating birds optimization algorithm.

B&B for the TSP

B&B algorithm guarantee optimality theoretically. The computational issue is a different story- meaning that the algorithms may fail to produce the optimum in a reasonable amount of time, prompting the development of heuristics.

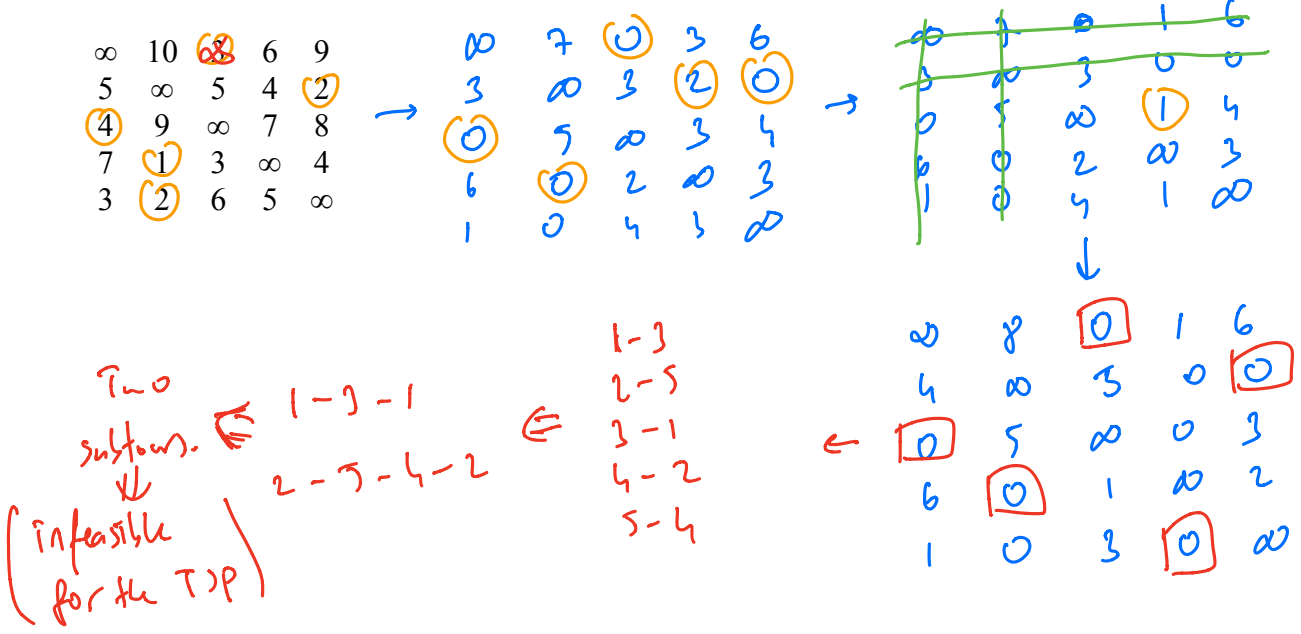
B&B starts with the optimum solution of the associated assignment problem.

If the solution is a tour, the process ends.

Otherwise, restrictions are imposed on the resulting solution to disallow subtours. The idea is to create branches that assign a zero value to each of the variables of one of the subtours. Normally, the subtour with the smallest number of cities is selected for branching because it creates the smallest number of branches.

Example:

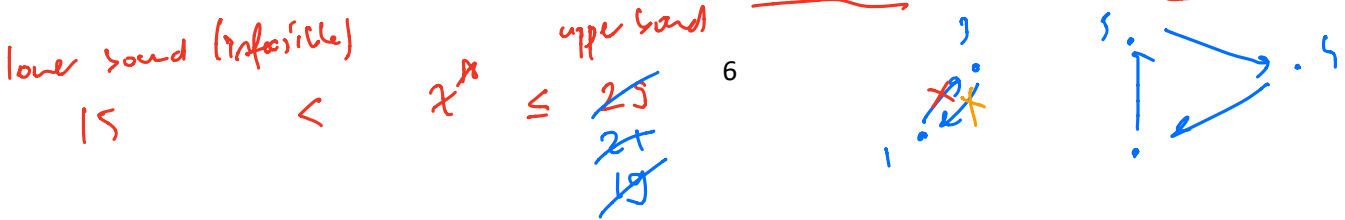
Consider the following 5-city TSP distance matrix:



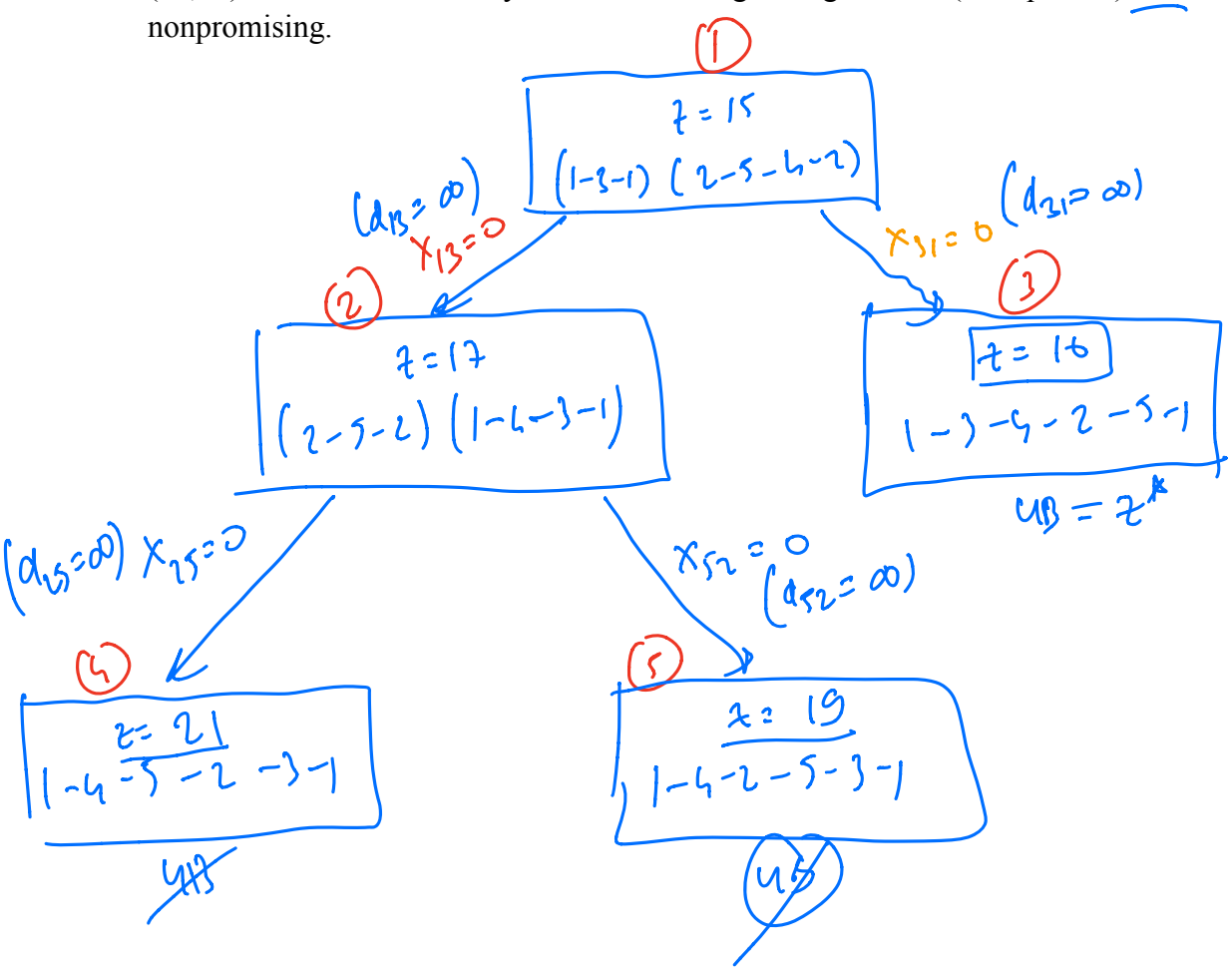
We start by solving the associated assignment, which yields the following solution:

$z=15$, $(x_{13}=x_{31}=1)$, $(x_{25}=x_{54}=x_{42}=1)$, all others.

This solution yields two subtours: (1-3-1),(2-5-4-2) with a total distance of 15, which is lower bound. A straightforward way to determine an upper bound is to select any tour and use its length as an upper bound estimate. For example, the tour 1-2-3-4-5-1 has a total length of 29.



The computed upper and lower bounds indicate that the optimum tour length lies in the range (15,29). A solution that yields a tour length larger than (or equal to) 29 is discarded as nonpromising.



To eliminate subtours at node 1, we need to disrupt its loop by forcing its member variables x_{ij} to be zero. So, we impose the restriction $x_{13}=0$ or $x_{31}=0$. (We consider this subtour for branching because it has fewer elements.) The associated assignment problems are created by setting $d_{13}=\infty$ and $d_{31}=\infty$.

Examine either node 2 or node 3.

Choose arbitrarily node 2.

Its assignment problem solution is 2-5-2 and 1-4-3-1 with z=17.

Solution is not a tour. Select 2-5-2 for branching:

Branch $x_{25}=0$ gives node 4 and branch $x_{52}=0$ gives node 5.

We now have three unexplored subproblems: nodes 3,4,5.

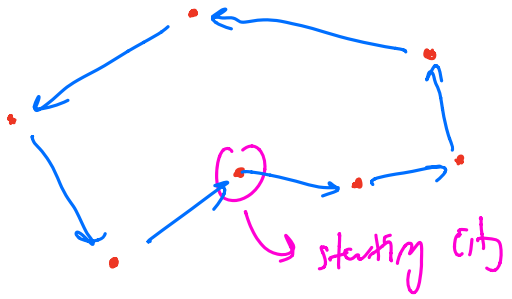
Arbitrarily examine node 4: Setting $d_{25}=\infty$. gives the solution 1-4-5-2-3-1 yields a smaller upper bound z=21.

Arbitrarily examine node 5: Setting $d_{52}=\infty$. gives the solution 1-4-2-5-3-1 with a smaller upper bound z=19.

Examine node 3: Setting $d_{31}=\infty$. gives the solution 1-3-4-2-5-1 with a smaller upper bound z=16.

All nodes are examined, thus completing the B&B search. The optimal tour is 1-3-4-2-5-1 with length of 16 units.

If we had examined node 3 before node 2, we would have encountered the upper bound z=16. So there would be no need to branch node 2 with z=17 because it cannot lead to better solution.



Local search heuristics

Nearest Neighbor

A TSP solution can be found by starting with a city and then connecting it to the closest unlinked city (break ties arbitrarily). The process continues until a tour is formed.

Ex.

	1	2	3	4	5
1	∞	120	220	150	210
2	120	∞	100	110	130
3	220	80	∞	160	185
4	150	∞	160	∞	190
5	210	130	185	∞	∞

3 - 2 - 4 - 1 - 5 - 3

80 110 150 210 185

735

Reversal Heuristic

In an n-city situation, the subtour reversal heuristic start with a feasible tour and then tries to improve on it by reversing 2-city subtours, followed by 3-city subtours and continuing until reaching subtours of size n-1.

Consider 1-3-5-2-4-1 in the following figure. Reversal of 3-5-2 produces the new tour 1-2-5-3-4-1.

